Placement-Robust Occupancy Detection with WiFi CSI, FMCW Radar, and LLM Reasoning

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***Abstract*—Occupancy detection is a basic smart home task. Wireless sensing is a private alternative to intrusive monitoring, but a new environment and a change of sensor placement both shift the signal statistics on which data-driven predictors depend, and machine-learning decisions arrive as black-box scores without a rationale. We develop an explainable occupancy detector that is calibrated per room from a few labeled empty and occupied sessions and then operates, without recalibration, at sensor placements unseen during calibration. We combine WiFi channel-state information (CSI) and frequency-modulated continuous-wave (FMCW) radar, extract physical descriptors and a three-level hierarchical PCA fitted only on calibration data, and supply descriptors and trained-model probabilities to large language models (LLMs) for a prediction and textual rationale. Calibration uses about six sessions per class at two placements (273 minutes, 86 occupied) for fitting and a third placement for thresholds; evaluation uses a fourth placement, where the receiver shares the CSI sender's wall. CSI alone does not survive the placement change (.793 validation, .365 test minute accuracy for its selected RF), whereas the radar RF ($.825/.804$), early-fusion RF ($.839/.811$), and late-fusion RF ($.922/.789$) transfer. GPT-4o mini reaches 88.2 ± 1.4% and GPT-5.4 76.6 ± 0.7% test accuracy; GPT-4o mini exceeds every validation-selected model while returning a rationale for each decision. The calibrated models are released as a pretrained starting point for new rooms.**

***Index Terms*—occupancy sensing, WiFi CSI, FMCW radar, domain shift, calibration**

I. Introduction

Room occupancy is a basic input for smart home automation, security, and in-home care. While camera-based occupancy detection raises privacy concerns, and wearables require cooperation or line of sight, radio frequency sensing offers a passive alternative: WiFi channel-state information (CSI) from commodity transceivers reflects multipath changes caused by a human body, and millimeter-wave frequency-modulated continuous-wave (FMCW) radar supplies range, angle, and Doppler maps that respond to body motion and even small movements like breathing.

Machine learning (ML) models outperform rule-based ones on wireless-sensing classification, but they remain unstable for continuous indoor monitoring tasks such as occupancy detection, localization, and activity recognition. We focus on the first and most fundamental of these, which enables applications including automations, elder care, and remote patient monitoring (RPM): an accurate, private, and reliable indoor occupancy detector. Wireless sensing limits privacy concerns; multiple modalities extend spatial coverage; and an LLM-based classifier partially addresses reliability by returning an intelligible decision and rationale from ML predictions and descriptors.

Both wireless sensing modalities, however, are sensitive to the environment and to the acquisition geometry. A new room changes the multipath environment entirely, so a deployed detector must be calibrated for it; within one room, moving a receiver to a different wall still changes the dominant multipath components, the radar field of view, and the CSI amplitude statistics on which a trained classifier depends. A model that separates the classes almost perfectly at the calibration placement can therefore fail at a new one. Machine-learning predictions also arrive without a rationale, which limits their use by operators who must decide whether to trust an alert.

We therefore adopt the two-stage workflow of Fig. 1: (i) *environment-specific calibration*, in which a pretrained occupancy model is adapted to a new room from a few short labeled empty and occupied sessions, and (ii) *placement robustness within that room*, in which the calibrated detector must keep working when the sensors move to positions never used for calibration. A practical advantage of this workflow is its low labeling cost: a handful of single-label sessions per class suffices, with no per-frame annotation. We study it on a self-collected multi-modal single-subject wireless sensing dataset whose calibration and evaluation placements differ (Fig. 2), and release the calibrated models as a pretrained starting point.[1] The main contribution is the first LLM-based wireless sensing occupancy detection pipeline: physical descriptors and three-level hierarchical training-basis PCA descriptors are calculated from CSI amplitude and FMCW radar; CSI-only, radar-only, early feature fusion, and late probability fusion are compared across machine learning approaches; and LLMs predict occupancy from a textual description of the problem and the descriptors, where the hierarchical PCA compresses high-dimensional sensor streams into a few components that fit the prompt while spanning enough time to reveal an occupancy-correlated pattern.

II. Related Work

A. WiFi CSI sensing

CSI exposes per-subcarrier channel gains and has been used for presence, activity, and gesture recognition with commodity hardware. DeMan [3] detects both moving and stationary humans by combining motion-induced CSI variation

[1] https://github.com/gadm21/radar

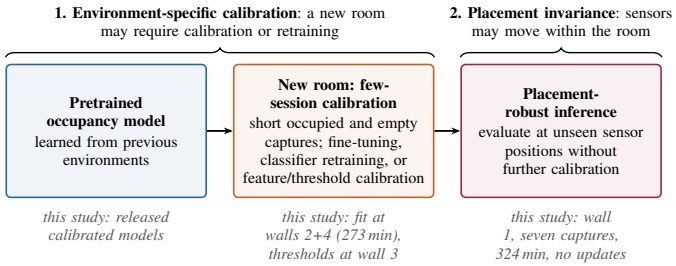
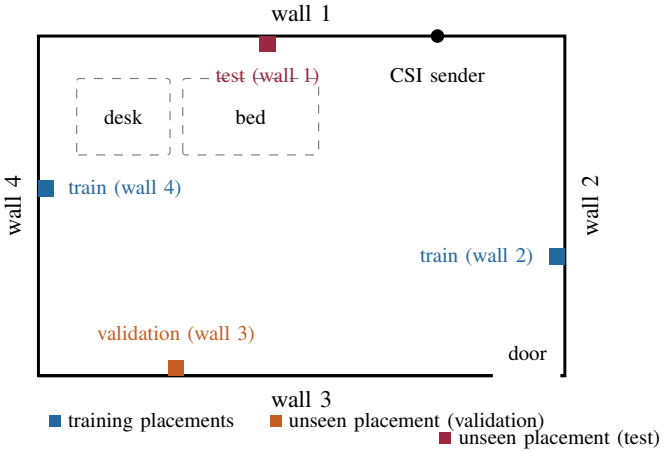
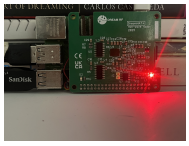


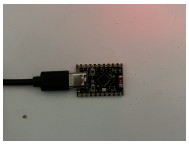
Fig. 1. Proposed occupancy training/deployment workflow: calibrate a pretrained model in a new room from a few labeled sessions, then evaluate at placements not used for calibration, with no further updates. Italic notes give this study's instantiation.

with respiration-induced periodicity, and reports that stationary presence is harder because the signal change is small.





radar: DreamHAT+ on Raspberry Pi



ESP32 board used for CSI

Fig. 2. Room layout, sensor placements, and collection hardware. The co-located FMCW radar and CSI receiver were mounted on wall 4 and wall 2 for training sessions, on wall 3 for validation sessions, and on wall 1 for test sessions; the CSI sender remained fixed on wall 1, so the test placement puts the receiver on the sender's own wall. Photographs show the DreamHAT+ mm-wave radar HAT mounted on its Raspberry Pi host (left) and the ESP32 board used for CSI collection (right).

TABLE I
FMCW RADAR CONFIGURATION. OPERATING BAND FROM THE BGT60TR13C PRODUCT BRIEF [1]; FRAME STRUCTURE FOLLOWS HOOD [2]

Configuration	Symbol	Value
FMCW radar (BGT60TR13C)		
Transmit / receive antennas	N_{Tx}/N_{Rx}	1 / 3
Chirps per frame / samples per chirp	N_c/N_s	64 / 128
Frame period / chirp interval	T_f/T_{cc}	50 ms / 391.55 μs
Start / stop frequency	$f_{\min}/f_{\max}$	58 / 63.5 GHz
Bandwidth	B	5.5 GHz

SHARP [4] targets environment- and person-independent activity recognition by estimating Doppler traces from CSI phase after sanitization, so that the learned representation depends less on static multipath. LiteHAR [5] replaces deep networks with random convolution kernels and a linear classifier. Our earlier work [6] processes amplitude from ESP32 access points for activity recognition in smart homes and motivates the amplitude-only CSI features used here. Placement shift remains a known weakness reported by these studies, and our multimodal result quantifies it for a fixed-threshold classifier.

In addition to CSI, other studies have explored FMCW radar for occupancy detection which has the advantage of not depending on a separate transmitter link. HOOD [2] trains a reconstruction-based detector on the Infineon BGT60TR13C and treats presence detection as an out-of-distribution problem. We use the same radar family but work from logged RD, RA, RE, and XY maps and compute low-dimensional descriptors (energy, temporal variability, peak-to-background) rather than training a reconstruction network, which keeps the models small enough to be summarized in a language-model prompt.

Chen *et al.* [7] fuse radar and WiFi for cross-scene activity recognition and report that fusion is more stable than either modality alone when the scene changes. We examine the same claim for a binary occupancy task with both early feature concatenation and late probability averaging. For model interpretation, SHAP [8] estimates each descriptor's contribution to a prediction; we use it to report which descriptors the frozen models rely on. Supplying model outputs and signal statistics to a large language model for a decision and a textual rationale is a newer practice, and we evaluate whether such rationales add information beyond the supplied model predictions.

III. Problem Definition

For each minute m, decide whether the room is empty ($y_m = 0$) or occupied ($y_m = 1$) from the CSI amplitude stream and the logged radar maps recorded in that minute. Calibration in the room uses the placements $\mathcal{C} = \{$walls 2, 4 (fitting); wall 3 (thresholds)$\}$: a classifier f is fitted on walls 2 and 4 and its cutoff τ on wall 3. Evaluation occurs at $p^* \notin \mathcal{C}$ (wall 1) without further calibration: no component may be updated with data from p^*. We compare the CSI, radar, early-fusion, and late-fusion views under this protocol and ask an LLM, given the descriptor statistics and frozen model outputs, to produce a decision and rationale measured against the same labels.

IV. System Model

Let $y_m \in \{0, 1\}$ denote empty or occupied in minute m. A CSI packet supplies a complex gain $h_i[k] = |h_i[k]|e^{j\phi_i[k]}$ on retained subcarrier k; we use the amplitude $|h_i[k]|$. Following HOOD [2], an FMCW chirp is $s(t) = \exp[j2\pi(f_c t + Bt^2/(2T_c))]$ for $0 < t < T_c$, with bandwidth B and chirp duration T_c; fast-time and chirp-axis FFTs yield range and Doppler. The available radar stream contains logged RD, RA, RE, and XY maps, from which we compute energy, temporal variability, and a peak-to-background proxy rather than HOOD's reconstruction score. The classifier returns $p_m =$

$P(y_m = 1 \mid x_{1,m}, x_{2,m})$. Early fusion concatenates CSI and radar descriptors before classification; late fusion averages separately fitted single-modal CSI and radar models' probabilities. Table I lists the radar configuration; the ESP32 CSI link uses one antenna per side, a 20 MHz channel, and 52 of 64 reported subcarriers on a 100 Hz grid, following [6].

V. Data Collection and Processing

A. Collection protocol

All data were collected from one subject in the single room shown in Fig. 2. The FMCW radar and the CSI receiver are co-located on one mount; the CSI sender is fixed on wall 1. Training sessions were recorded with the mount on wall 2 and on wall 4. Validation sessions were recorded with the mount on wall 3, a placement absent from training, in two collection rounds: single-link sessions and four multilink captures whose extra links are unused here (they retain their capture identity for minute nesting). Test sessions were recorded later with the mount on wall 1, a second unseen placement on the CSI sender's own wall. Each session runs the multimodal collection (CSI and radar logged together) for a fixed duration under one condition: the subject is present in the room and the session is labeled *occupied*, or the subject leaves and the session is labeled *empty*. Labels are therefore assigned per session, not per frame, and no session contains an occupancy transition.

B. Data sets and preprocessing

Table II and Fig. 3 summarize the data sets. We measure the data duration in minutes and fifteen-second windows (since the raw input length for the hierarchical PCA is fifteen seconds). The development collection provides 273 recorded training minutes (187 empty, 86 occupied). Because occupied is the minority class in the development sets, we synthesize occupied minutes by perturbing the recorded occupied minutes at the descriptor level: Gaussian noise, gain drift, mixup between occupied minutes, and temporal resampling, in rotation. This adds 101 synthetic training minutes, giving a balanced set of 374 minutes (1,496 fifteen-second windows). The validation collection combines 192 single-link wall-3 minutes (112 empty, 80 occupied) with 360 wall-3 multilink capture minutes (180 per class), totalling 552 recorded minutes (292 empty, 260 occupied) plus 32 synthetic occupied minutes, i.e., 584 minutes (2,336 windows). Seven wall-1 test captures (two empty, five occupied) provide 1,301 fifteen-second windows and 324 complete minute buckets (111 empty, 213 occupied) and are never augmented. Each capture has one label, so minutes are nested within captures. An occupied-only calibration session recorded at the training placements is excluded because it cannot inform a two-class rule. Data labeled as occupied include different activities from the subject present in the room, including sleeping, sitting, walking, and standing.

C. Physical and hierarchical PCA descriptors

Candidates include CSI amplitude/variance, RD/RA/RE/XY map summaries, and an SNR proxy: $10 \log_{10}$ of nonmasked RD peak over median RD power. Figure 4 shows the RA and RE

TABLE II
Data sets: recorded minutes per class (with synthetic occupied minutes added for class balance) and collection dates, all in 2026.

Set	Minutes per class	Collection dates
Training	187 empty / 86 occupied (+101 synthetic)	September 6–8
Validation	292 empty / 260 occupied (+32 synthetic)	September 15; October 3–5
Test	111 empty / 213 occupied	October 19–20

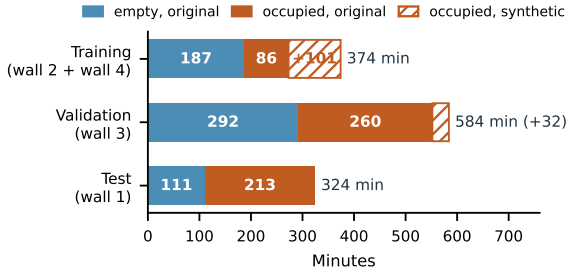


Fig. 3. Data sizes in minutes per class: recorded (solid) and synthesized occupied minutes added for class balance (hatched). Test captures are never augmented.

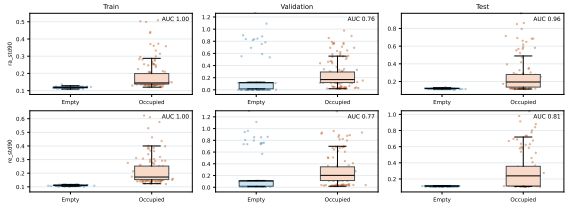


Fig. 4. RA and RE variability by class in the training, combined validation, and wall-1 test sets.

descriptors. Each modality also carries a three-level hierarchical PCA. Level 1 projects one-second CSI dots (mean/std of the 52 subcarrier amplitudes) and radar dots (flattened mean RD map) onto a two-component basis fitted on training seconds. Level 2 stacks five consecutive level-1 pairs (10 values) and level 3 stacks three consecutive level-2 pairs (6 values), each fitted by a two-component PCA on the corresponding training blocks, so every level is fitted on the training output of the level below and validation and test seconds are only projected. Blocks with fewer than two valid seconds and windows with fewer than two valid blocks yield missing values that receive training medians (24% of training and 10% of validation hierarchical values; no test value is missing). Each fifteen-second window summarises the hierarchy by the mean and standard deviation of its level-1 and level-2 PC1/PC2 coordinates and its level-3 PC1/PC2 coordinates (ten values per modality). Model inputs below always combine the physical descriptors with these hierarchical PCA descriptors; Fig. 5 shows the three levels for test data.

VI. Predictive Models and Evaluation

We fit RF, GBM, SVM, and a two-hidden-layer MLP (64/32 units) on CSI, radar, and their early descriptor concatenation,

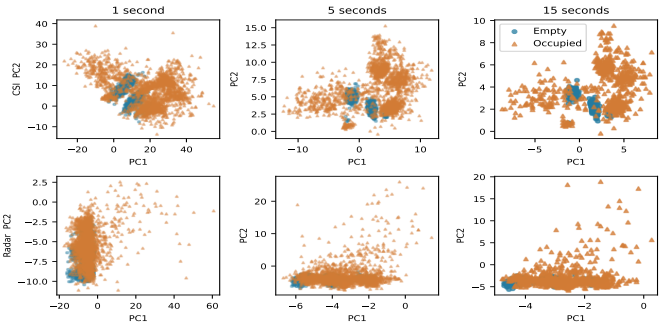


Fig. 5. Hierarchical PCA coordinates (PC1 vs PC2) at the one-second, five-second, and fifteen-second levels for CSI (top) and radar (bottom); wall-1 test points projected through training-fitted bases, no test fitting.

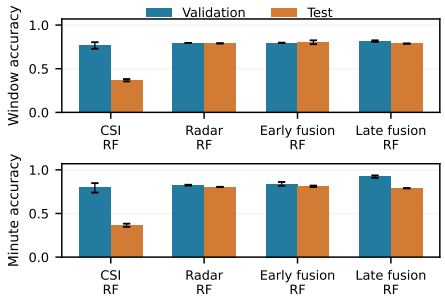


Fig. 6. Window and minute accuracy on the combined validation (blue) and wall-1 test (orange), using validation-fixed models and thresholds.

each with five random seeds. Late fusion averages same-family CSI and radar probabilities with fixed equal weights. The balanced training set fits missing-value medians, scalers, and model weights; the combined validation set fixes each seed's cutoff by balanced accuracy, and mean validation-minute AUC over the five seeds selects each view's model and the conventional (ML) and neural (DL) reference models. All choices precede wall-1 test evaluation. Minute probabilities average four fifteen-second outputs; window comparisons use the same minute-fitted cutoff. Test data update no model component.

Classifier vectors contain 18 CSI inputs (eight physical descriptors averaged over the window's three five-second rows: amplitude mean/std/P90; rolling-variance mean/P90/max; temporal variance; Doppler fraction; plus the ten hierarchical PCA descriptors) and 32 radar inputs (SNR mean/max; mean, P90, variability, frame delta, and peak of each RD, RA, RE, and XY map; plus ten hierarchical PCA descriptors), giving 50 early-fusion inputs. Each view has training-fitted medians and z scores. Validation fixes the late-fusion RF (ML reference) and radar MLP (DL reference) cutoffs at .622 and .978; Fig. 7 shows their frozen path.

VII. Language-Model Inference

Each LLM receives 29 minute means: CSI amplitude mean/std and rolling variance; radar SNR, RA/RE/XY variability 90th percentiles, RA/RE frame deltas; and the twenty hierarchical PCA descriptors of both modalities. One request combines a system rule with a JSON user body that adds training-class mean/std, two labeled training-minute examples per class, and each reference's validation-fixed cutoff, balanced accuracy, minute probability, and decision (Fig. 7). The system rule asks the LLM to respect model agreement and compare cutoff margins with descriptor statistics on disagreement. IDs are shuffled; capture names, labels, and test-batch ranks are excluded. Output probabilities use a fixed .5 decision cutoff. Three runs cover all 324 wall-1 test minutes with the same frozen prompt.

VIII. Results and Discussion

A. CSI-only results

Validation selects the CSI RF (AUC .81, accuracy .79), but on the wall-1 placement it collapses to .37 ± .02 test minute accuracy (window .37, AUC .60 ± .12): seed 0 labels nearly every minute empty, correctly rejecting all 111 empty minutes while missing 207 of 213 occupied ones (Table IV). The CSI GBM (.392), SVM (.343), and MLP (.312) fail similarly. Mounting the receiver on the sender's own wall removes the cross-room path on which the CSI descriptors depend, so no CSI model transfers.

B. Radar-only results

The radar RF leads its view on validation (AUC .92, accuracy .83) and transfers well: .804 test minute accuracy (window .790, AUC .992) with no empty-minute errors and 64 missed occupied minutes, most inside one difficult occupied capture. The radar GBM (.799), SVM (.791), and MLP (.75±.09) score comparably, the MLP being the DL reference because it is the strongest MLP view on validation.

C. Sensor fusion: early features

Early fusion concatenates the 50 inputs. The RF leader (validation AUC .93, accuracy .84) reaches .811 ± .01 test minute accuracy (window .803, AUC .993), inheriting the radar view's error pattern. The early-fusion GBM scores .969 on test but ranks below the RF on validation AUC (.87), so it is not selectable.

D. Decision fusion: late probabilities

Late fusion averages same-family CSI and radar probabilities with weight $1/2$ fixed before evaluation. The RF pair posts the highest validation AUC of all sixteen candidates (.97, accuracy .92) and becomes the ML reference; on test it reaches .789 minute accuracy (window .788, AUC .978), held back by the CSI leg's collapse. The late-fusion GBM reaches .972 test with 9 empty-minute and no occupied-minute errors, although its validation AUC (.85) keeps it from being selected.

E. Descriptor importance

Table V reports interventional Shapley estimates for the CSI RF, the early-fusion MLP, and the late-fusion GBM. Each of 24 random descriptor orders starts from a random training window; 24 validation windows (12 per class) are explained. The CSI RF relies on CSI amplitude P90 (.157), std (.074),

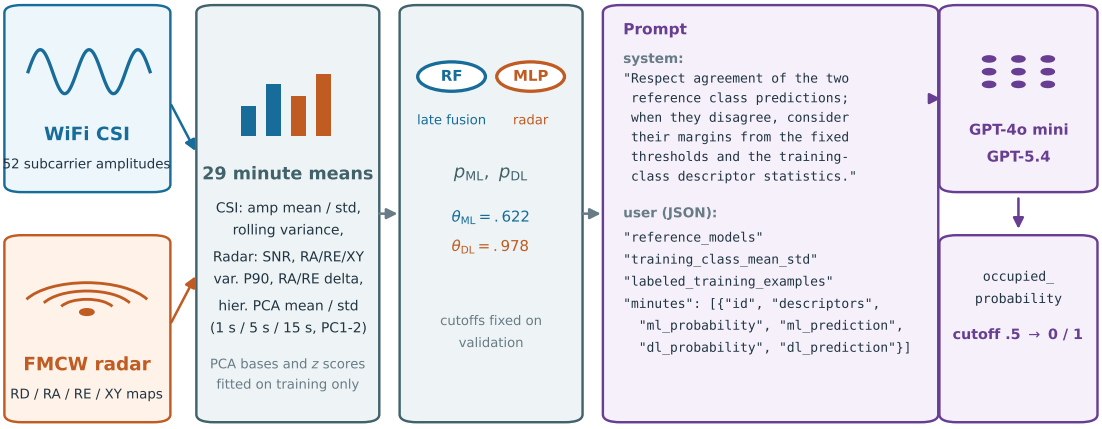


Fig. 7. Frozen descriptor-to-LLM path. Twenty-nine minute-mean descriptors and the late-fusion RF and radar MLP reference probabilities with their validation-fixed cutoffs enter a single chat request; the card shows the system rule and the JSON user keys. The LLM returns an occupancy probability thresholded at .5.

TABLE III
SIX WALL-1 TEST MINUTES (THREE PER CLASS). A/V: CSI AMPLITUDE/ROLLING-VARIANCE MEAN; S: RADAR SNR; $R_A/R_E/X$: RADAR VARIABILITY. REFERENCES ARE LATE-FUSION RF / RADAR MLP; LLM REASONS ARE VERBATIM

ID/actual	Signal summary	Ref. p	GPT-4o mini: prediction, reason	GPT-5.4: prediction, reason
m0096/E	CSI A/V: 21.71/0.070 Radar S: 7.38 $R_A/R_E/X$: 0.12/0.11/0.11	0.41/ 0.00	**E (0.41)**. CSI amplitude and radar features align with empty class; ML probability below threshold.	**E (0.41)**. CSI motion metrics look empty-like despite extreme PC shifts; radar PCs strongly non-occupied; RF-fusion 0.414 empty, radar-MLP 0.004 empty.
m0235/E	CSI A/V: 21.84/0.073 Radar S: 7.50 $R_A/R_E/X$: 0.12/0.11/0.11	0.37/ 0.02	**E (0.37)**. Similar CSI and radar features to empty examples; low ML and DL probabilities.	**E (0.37)**. CSI ra/re/xy spreads remain empty-like though PC means are unusual; radar PCs are strongly negative empty-like; RF-fusion 0.370 empty, radar-MLP 0.015 empty.
m0155/E	CSI A/V: 25.02/0.042 Radar S: 7.51 $R_A/R_E/X$: 0.12/0.11/0.11	0.38/ 0.00	**E (0.38)**. CSI and radar features suggest empty; both model probabilities below thresholds.	**E (0.38)**. CSI spreads and deltas are empty-like despite very large PCA offsets; radar PCs strongly negative empty-like; RF-fusion 0.384 empty, radar-MLP 0.002 empty.
m0195/O	CSI A/V: 28.46/0.049 Radar S: 7.57 $R_A/R_E/X$: 0.27/0.27/0.14	0.72/ 1.00	**O (0.72)**. High CSI amplitude and radar features consistent with occupied class; both model probabilities above thresholds.	**O (0.72)**. CSI ra/re spreads exceed empty and approach occupied; radar remains atypically negative, conflicting; RF-fusion 0.723 occupied, radar-MLP 1.000 occupied.
m0162/O	CSI A/V: 26.18/0.053 Radar S: 7.83 $R_A/R_E/X$: 0.21/0.21/0.15	0.59/ 0.84	**E (0.59)**. Conflicting evidence; high CSI but ML prediction indicates empty.	**E (0.59)**. CSI ra/re/xy and deltas are occupied-like with huge PC offsets, but radar PCs stay negative empty-like; RF-fusion 0.587 empty, radar-MLP 0.841 empty.
m0041/O	CSI A/V: 28.59/0.053 Radar S: 7.90 $R_A/R_E/X$: 0.43/0.52/0.21	0.71/ 1.00	**O (0.71)**. Strong CSI and radar evidence for occupied; both model probabilities confirm.	**O (0.71)**. CSI ra/re/xy spreads and deltas are strongly occupied-like; radar PCs still negative so conflict remains; RF-fusion 0.707 occupied, radar-MLP 1.000 occupied.

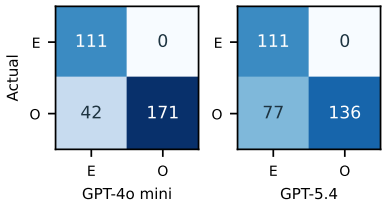


Fig. 8. Test confusion matrices for the two LLMs over the 324 wall-1 minutes, decision at .5 on the mean of three runs. E/O: empty/occupied; rows are actual, columns predicted.

and mean (.049), followed by total variation (.026) and the five-second and one-second PC1 means (.024 each). The early-fusion MLP concentrates on RA energy mean (.432) and P90 (.413), CSI amplitude mean (.208), and RE energy mean (.176). The late-fusion GBM uses CSI amplitude P90 (.219), RE variability (.132), CSI amplitude mean (.095), and RE frame delta (.081); its largest hierarchical term is the one-second CSI PC1 mean (.032). These values describe model reliance, not causal sensor importance [8].

F. Language-model evaluation

GPT-4o mini scores $88.2 \pm 1.4\%$ on the wall-1 test set and GPT-5.4 scores $76.6 \pm 0.7\%$ (Fig. 8). Both models reject every empty minute; all errors are missed occupied minutes, most in the capture that also defeats the radar leg (31 of 42 for GPT-4o mini, 61 of 77 for GPT-5.4). GPT-4o mini thus exceeds every validation-selected leader (.811); GPT-5.4 stays just below. On that capture the ML reference votes occupied in only 10% of minutes; GPT-5.4 tracks the reference probability more closely (correlation .94 versus .82), so it inherits the references' error, while excluding that capture the two LLMs score .957 and .937.

TABLE IV

FIFTEEN-SECOND MODELS ON PHYSICAL + HIERARCHICAL PCA DESCRIPTORS, MINUTE EVALUATION, THRESHOLDS FIXED ON THE COMBINED VALIDATION; MEAN ± SD OVER FIVE SEEDS (SD OMITTED WHEN $< .005$). * VIEW LEADER BY MEAN VALIDATION-MINUTE AUC

View	Model	Val. AUC	Val. acc.	Test acc.	Test AUC
CSI	RF*	.81±.04	.79±.05	.36±.02	.60±.12
	GBM	.80	.78	.39	.87
	SVM	.39	.59	.34	.82
	MLP	.60±.04	.65±.04	.31±.07	.14±.03
Radar	RF*	.92±.00	.82±.00	.80±.00	.99±.00
	GBM	.83	.77	.80	1.00
	SVM	.88±.00	.80±.00	.79±.00	.88±.00
	MLP	.82±.05	.80±.03	.75±.09	1.00±.00
Early fusion	RF*	.93±.02	.84±.02	.81±.01	.99±.01
	GBM	.87	.81	.97	1.00
	SVM	.83±.00	.74±.00	.80	.87
	MLP	.78±.08	.79±.04	.54±.04	.84±.19
Late fusion	RF*	.97±.01	.92±.01	.79±.00	.98±.03
	GBM	.85	.78	.97	1.00
	SVM	.86	.80	.76±.01	.90±.00
	MLP	.72±.04	.74±.04	.42±.16	1.00±.00

TABLE V

MEAN ABSOLUTE INTERVENTIONAL SHAPLEY CONTRIBUTION TO OCCUPIED PROBABILITY FOR THE CSI RF (C-RF), THE EARLY-FUSION MLP (E-MLP), AND THE LATE-FUSION GBM (L-GBM); 24 VALIDATION WINDOWS, TRAINING-WINDOW BACKGROUND

Descriptor	C-RF	E-MLP	L-GBM	Descriptor	C-RF	E-MLP	L-GBM
CSI amplitude mean	.049	.208	.095	RA energy mean	–	.432	.001
CSI amplitude std	.074	.007	.020	RA energy P90	–	.413	.000
CSI amplitude P90	.157	.126	.219	RA variability P90	–	.023	.001
CSI rolling-var. mean	.011	.004	.004	RA frame delta	–	.091	.000
CSI rolling-var. P90	.007	.004	.004	RA peak/bkg.	–	.088	.006
CSI rolling-var. max	.003	.006	.005	RE energy mean	–	.176	.000
CSI total variation	.026	.007	.014	RE energy P90	–	.128	.003
CSI Doppler fraction	.013	.005	.008	RE variability P90	–	.032	.132
CSI 1 s PC1 mean	.024	.095	.032	RE frame delta	–	.071	.081
CSI 1 s PC2 mean	.005	.081	.014	RE peak/bkg.	–	.128	.000
CSI 1 s PC1 std	.015	.007	.008	XY energy mean	–	.042	.000
CSI 1 s PC2 std	.009	.015	.003	XY energy P90	–	.045	.000
CSI 5 s PC1 mean	.024	.026	.006	XY variability P90	–	.080	.011
CSI 5 s PC2 mean	.006	.003	.007	XY frame delta	–	.096	.000
CSI 5 s PC1 std	.015	.005	.011	XY peak/bkg.	–	.034	.002
CSI 5 s PC2 std	.015	.011	.002	Radar 1 s PC1 mean	–	.004	.000
CSI 15 s PC1	.011	.080	.013	Radar 1 s PC2 mean	–	.048	.000
CSI 15 s PC2	.007	.009	.004	Radar 1 s PC1 std	–	.003	.000
Radar SNR mean	–	.003	.000	Radar 1 s PC2 std	–	.006	.000
Radar SNR max	–	.002	.000	Radar 5 s PC1 mean	–	.032	.000
RD energy mean	–	.002	.000	Radar 5 s PC2 mean	–	.009	.000
RD energy P90	–	.009	.000	Radar 5 s PC1 std	–	.035	.000
RD variability P90	–	.002	.000	Radar 5 s PC2 std	–	.002	.000
RD frame delta	–	.011	.000	Radar 15 s PC1	–	.030	.000
RD peak/bkg.	–	.047	.001	Radar 15 s PC2	–	.013	.000

G. LLM explanation check

For a qualitative check, three empty and three occupied test minutes were sampled with a fixed seed, without considering model outcomes. Both LLMs received the same inputs as above with a request for a short reason (a separate prompt, excluded from accuracy); Table III reports their verbatim replies. Both models resolve the three empty minutes and two of the occupied ones; their single shared miss is an occupied minute on which both references already vote empty. In this prompt both models report the RF reference probability as their own. These are model-generated rationales, not causal explanations of raw signals.

IX. LIMITATIONS

Labels are constant within each capture, so adjacent minutes are correlated, occupancy transitions are not measured, and test replication is two empty and five occupied captures. The unseen domain is one held-out placement for one subject in one room. By design a new room requires calibration; the claim is placement robustness after it, and it holds for radar-based views and the LLMs but not for CSI alone. The 32 synthetic occupied minutes in the validation set influence threshold choice. LLM hallucination also limits the explainability claim: a plausible rationale is not an independent explanation of the measurement.

X. CONCLUSION

We presented a privacy-preserving occupancy workflow that calibrates a WiFi CSI and FMCW radar detector once per room from a few labeled sessions—physical descriptors together with a three-level hierarchical PCA fitted only on calibration data—and then evaluated it, with all models and thresholds frozen, at a sensor placement never used for calibration. There the CSI view collapses (.365 test for its validation-selected RF), radar-based views transfer (.804–.811 for the selectable radar and early-fusion RFs), and late probability fusion keeps the highest validation AUC (.922 validation, .789 test), confirming that a placement change can be catastrophic for one modality yet mild for another. Supplying descriptor statistics and frozen model outputs to LLMs yielded $88.2 \pm 1.4\%$ test accuracy for GPT-4o mini and $76.6 \pm 0.7\%$ for GPT-5.4; the former exceeds every validation-selected machine-learning view while returning a textual rationale for each decision, though LLM accuracy remains sensitive to the supplied references. For radar-based views and the LLMs, moving the sensors required no recalibration.

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